

From Dynamics to Design: Computational Insights into Fluorogenic RNA Aptamers

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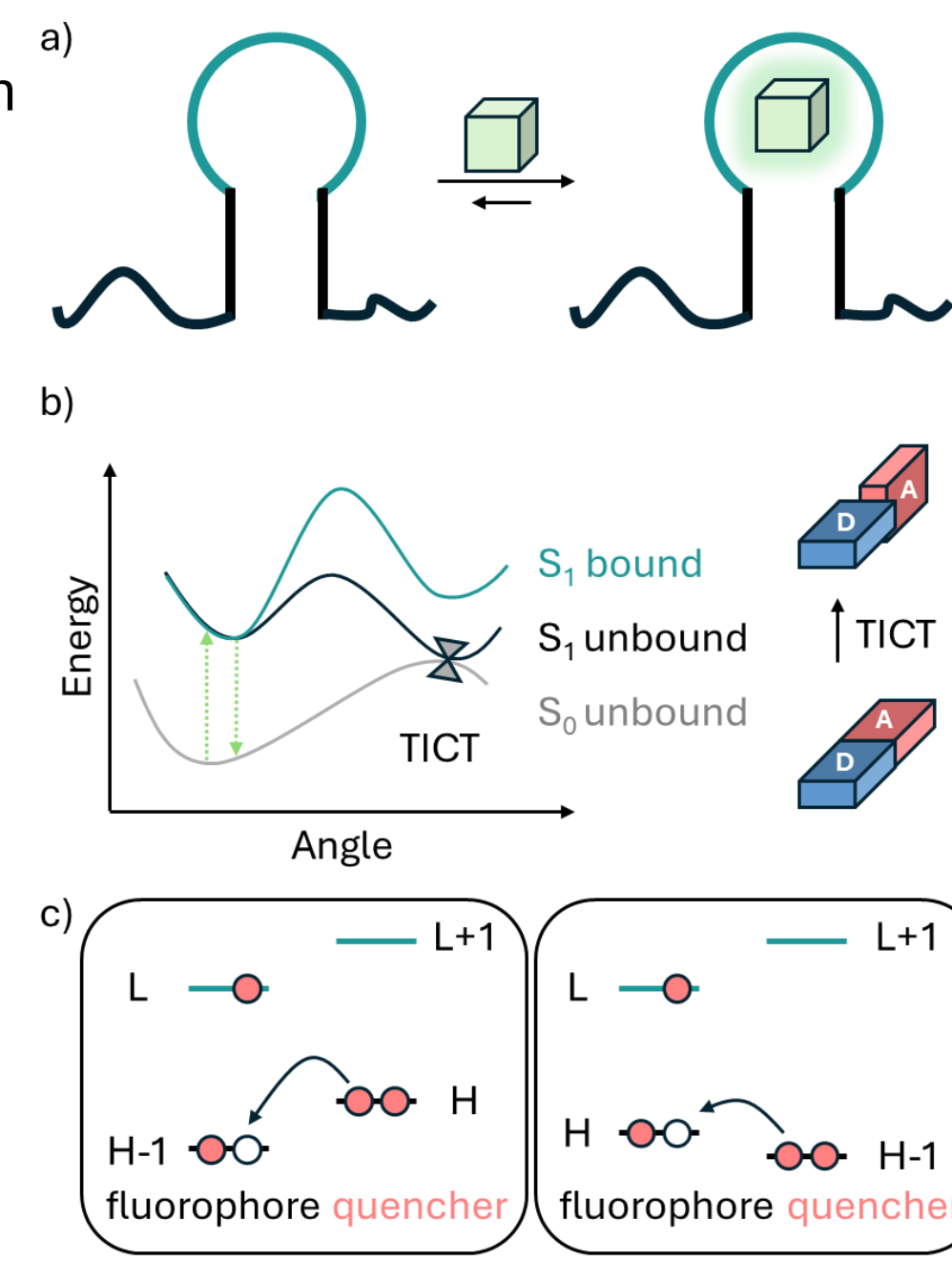
Introduction

Genetically encoded fluorescent RNAs are powerful tools for visualizing RNA dynamics and enabling the development of RNA-based biosensors. Unlike fluorescent proteins, which rely on an intrinsic, covalently embedded chromophore, fluorogenic RNA aptamers use small-molecule fluorophores that bind through non-covalent interactions. This modularity allows for independent engineering of both the RNA and its fluorogenic ligand, opening the door to finely tuned optical and functional properties.¹

Inspired by the conditional fluorescence of the green fluorescent protein (GFP) chromophore *p*-HBI, the Jaffrey lab pioneered the field in 2011 by synthesizing fluorogenic analogs of HBI and selecting aptamers that activate fluorescence upon binding (a).² Using an *in vitro* selection method based on competitive binding, they identified RNA-fluorophore pairs with diverse emission profiles across the visible spectrum. Since then, an array of *fruit and veggie* RNA aptamers, including Broccoli, Corn, Squash, Okra, Pepper, Chili, Mango, and Beetroot, have been developed. To date, light-up RNA aptamer systems exploit two general mechanisms of fluorogenicity: twisted intramolecular charge transfer (TICT) (b) and contact quenching (CQ) (c) leading to a high diversity of fluorescence and binding properties.

In TICT, upon excitation, the fluorophore adopts a twisted conformation that stabilizes a charge-separated state often resulting in fluorescence quenching depending on the rigidity of the environment. RNA binding can restrict this twisting, thereby enhancing fluorescence (b). CQ, on the other hand, occurs if the energy level of the highest occupied molecular orbital (H) of the quencher fragment is higher than that of the fluorophore (c). In some cases, if the H-1 of the quencher is close enough in energy to the H of the fluorophore, quenching can occur as well. Despite the progress in synthesizing several HBI-based aptamer systems for various applications, the structure-function relationship between modified HBI derivatives and different aptamers remains poorly understood.

Herein, we investigate HBI derivatives in the Spinach, Broccoli, and Squash aptamer systems using a multifaceted computational approach to elucidate their binding interactions and nonradiative decay mechanisms.



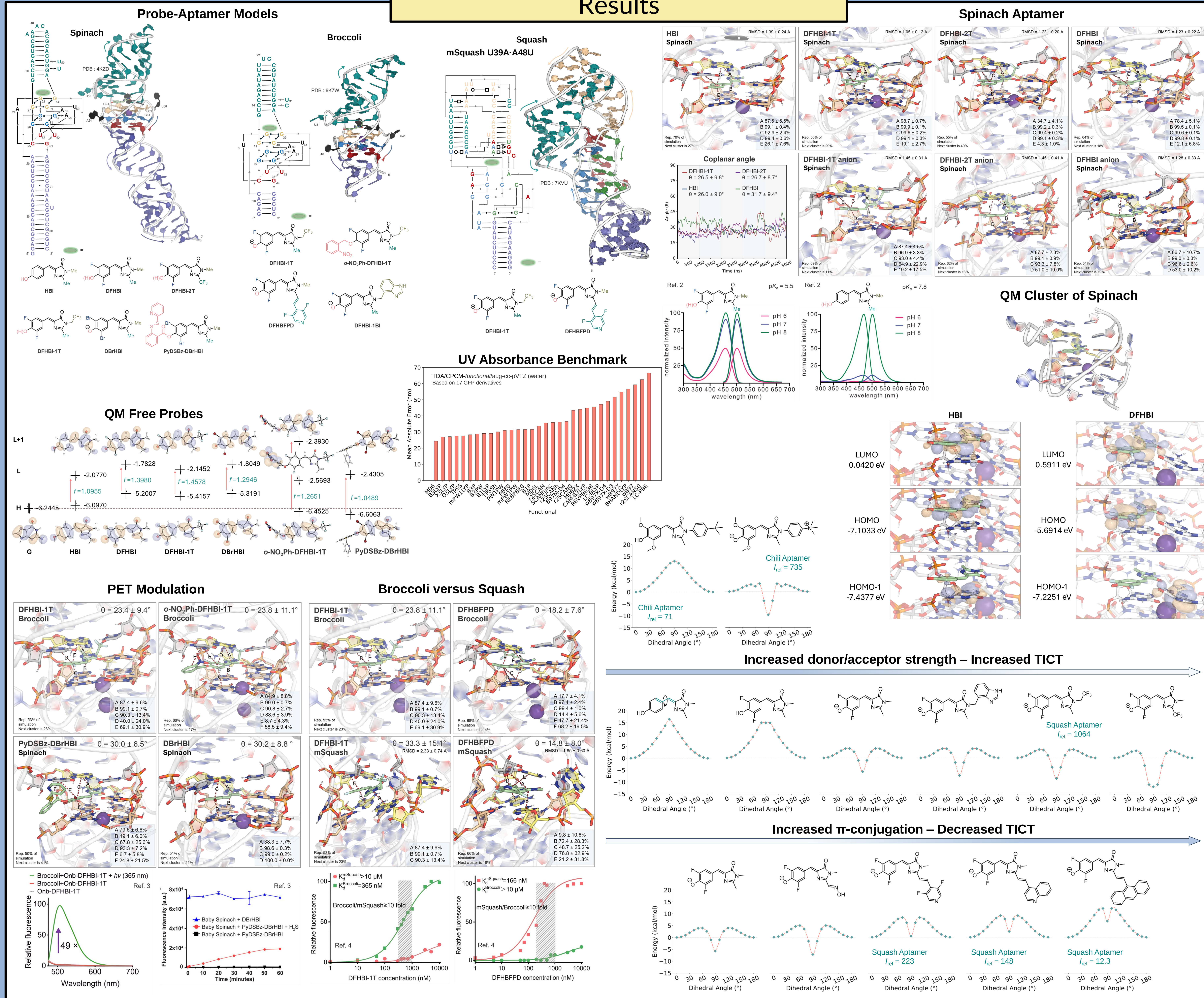
Computational Methods

Crystal structures of Spinach, Broccoli, and Squash were used to build all models (PDB ID: 4KZD⁶, 8K7W⁷, and 7KVU⁸, respectively). MD simulations used the AMBER RNA (OL3), water (TIP4P-EW), and general amber (GAFF) force fields. The models were solvated in a rectangular water box with a 12 Å distance from solute. The system was neutralized with a KCl concentration of 150 mM and MgCl₂ concentration of 10 mM. Four steps of minimization, six steps of heating, and five steps of equilibration were carried out prior to production. Constant molar, pressure and temperature (NVT) ensemble MD simulations were run using AMBER20 for 1000 ns in quintuplicate for each system. The excited state geometries for the rotational electronic energy profiles of free probes were optimized using TD-DFT/LR-PCM- ω B97X-D3/def2-SVP (water) with Gaussian 16 (rev. C.01). Single-point energy calculations were carried out using TD-DFT/cLR-SMD- ω B97X-D3/def2-SVP (water). The ground state geometries of free probes were optimized using PCM- ω B97X-D3/def2-SVP (water) with ORCA 6.0.1. Excited state single-point energy calculations were carried out using TDA/CPCM-M06/aug-cc-pVTZ (water) which benchmarking proved to be the most robust. Hierarchical agglomerative clustering on 500 equidistant structures over 5000 ns of MD simulation were used to obtain probe-aptamer clusters which was further optimized with ALPB-GFN2-xTB-FF (water) with ORCA 6.0.1. Excited state single-point energy calculations were carried out using TDA/CPCM- ω B97X-D3/def2-SVP (water). Ground state single-point energy calculations were carried out using CPCM- ω B97X-D3/def2-SVP (water). All calculations carried out with ORCA 6.0.1 were accelerated using the resolution of identity approximation (RI) for the Coulomb integrals, while the exchange terms were efficiently computed using the 'chain-of-spheres' (COSX) approximation with the def2/J coulomb fitting auxiliary basis set.

Conclusions

Structurally, most fluorogenic probes can fit within the binding pockets of various aptamers with reasonable compatibility. Once intercalated, the dynamic flexibility of the probe does not appear to be the primary factor influencing fluorescence output in most cases. Instead, the binding and dissociation rate constants are likely more critical in determining the observed fluorescence turn-on response than simple restriction of TICT motion within the pocket. Aptamers that contain G-quadruplex motifs are particularly susceptible to quenching via photoinduced electron transfer (PET) when paired with electron-rich (neutral) probes, leading to reduced fluorescence enhancement. This explains the relatively low fluorescence observed in systems like Spinach and Broccoli when neutral probes are used. Neutral GFP-like fluorophores are especially prone to PET quenching in the aptamer environment. For example, the neutral HBI generates a photoinduced hole slightly higher in energy than the HOMO of guanine, enabling PET quenching upon binding to guanine-rich motifs. As a result, fluorescence can be effectively eliminated in the neutral form. Upon deprotonation, however, the HOMO energy level of HBI is raised, suppressing PET and turning fluorescence back on. Other probes such as *o*-NO₂-Ph-DfHBI-1T undergo a HOMO-1 to LUMO transition that may lead to self-quenching through PET, even in the absence of nearby guanines. Similarly, PyDSBz-DBrHBI is PET-quenched by guanine until it is converted to anionic DBrHBI, at which point fluorescence can be restored. In addition to PET, control over twisted intramolecular charge transfer (TICT) offers another layer of tunability. Minor electronic modifications, such as increasing the electron-donating strength of donor groups, enhancing the electron-withdrawing nature of acceptors, or reducing π -conjugation, can modulate the TICT propensity. These changes influence fluorescence by promoting or suppressing nonradiative decay pathways in the unbound state, providing a strategic method to minimize unwanted background fluorescence. Taken together, these findings provide a mechanistic basis for optimizing fluorogenic response through structural tuning of both aptamers and their ligands. They offer a roadmap for engineering high-performance fluorescent probes tailored to specific nucleic acid targets. Ultimately, the modularity of both components supports a synthetic biology paradigm in which fluorescence output is predictably programmable through molecular design.

Results



References

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